

RSR WEEK 08

1. Which R formula correctly fits a multiple linear regression predicting sick_days fi and kids?

a) `lm(washes ~ sick_days + kids, data = handwashing)`

b) `lm(sick_days + washes ~ kids, data = handwashing)`

c) `lm(sick_days ~ washes + kids, data = handwashing)` ✓

2. In a multiple linear regression, each slope is the change in the outcome for a one in that predictor while:

a) Averaging the slope across all possible interaction terms

b) Setting every other predictor to its sample maximum

c) Holding the other predictors in the model constant ✓

3. [SEE IMAGE] Two predictors in the same model: one has Estimate = 0.05 (per minute) and the other Estimate = 4.2 (per item). Why can you NOT conclude the second predictor 'n

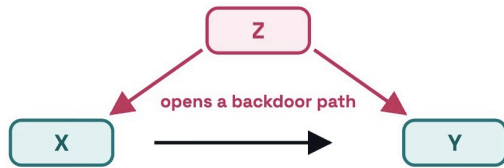
```
lm(Y ~ minutes + count, data = dat)
              Estimate
(Intercept)    8.00
minutes         0.05 # per minute
count          4.20 # per item
```

a) Because the predictors are on different units / scales, so raw slopes are not comparable head-to-head ✓

b) Because only standardized coefficients are ever interpretable

c) Because the larger estimate always has the larger p-value

4. [SEE IMAGE] A third variable influences both the predictor of interest and the outcome variable, that third variable is called a:

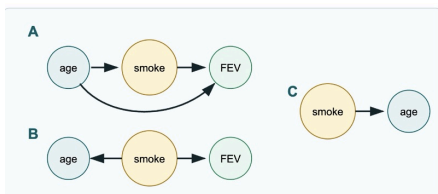


a) A residual (the leftover error after fitting the model)

b) A mediator (it sits on the causal path from X to Y)

c) A confounder (it creates a non-causal association between X and Y) ✓

5. [SEE IMAGE] Which DAG shows confounding by age?



a) C. Smoking causes age

b) B. Smoking points to age and FEV

c) A. Age points to smoking and to FEV

